Constraint-Aware DNA Encoding for Semantic Healthcare Data Archival

Sanket Patil
Department of Information Technology
Vivekanand Education Society's Institute of Technology
Mumbai, India
2023.sanket.patil@ves.ac.in

Dr.Manoj Sabnis
Department of Information Technology
Vivekanand Education Society's Institute of Technology
Mumbai, India
manoj.sabnis@ves.ac.in

Mrs.Kajal Joseph
Department of Information Technology
Vivekanand Education Society's Institute of Technology
Mumbai, India
kajal.joseph@ves.ac.in

Abstract.

In India alone, thousands of clinical records are recorded daily in its numerous clinics and diagnostic centers; however, they are stored in unstructured formats such as unencrypted PDFs which do not offer any compression nor indexing nor preservation strategy. While DNA storage is an approach offering high physical density and durability, virtually all prior works on the topic perform input encoding of their respective information systems as an opaque bitstream, disregarding any potential semantic information associated with it. In this paper, we propose the design of Project Helix, a DNA data archiving framework specifically designed for clinical data, including record conversion to a representation of schemas, followed by normalization in view of semantics specific to clinical domain, and then compression using run-length, Huffman, and arithmetic encoding, and finally mapping the encoded stream onto synthetic DNA sequences subject to biological constraints. The architecture behind each step of the proposed pipeline, the encoding algorithm considering biological constraints, and our feasibility experiments with an implemented prototype on real clinical data gathered by our collaborators in Homoeopathy and Ayurveda are presented. Compression ratio of 2.53× was achieved using only classical and arithmetic coding techniques relative to structured binary encoding, while the generated DNA strings had 48.1% average GC content and homopolymer run lengths not exceeding two nucleotides – all within the bounds acceptable for biological synthesis. It is important to emphasize that this paper focuses on the architecture and encoding aspect of the presented framework, and the AI-based latent compression and decoding layer constitutes a natural extension of this framework, but is studied separately.

Keywords: DNA data archiving, healthcare data archiving, semantic compression, constraint-aware encoding, medical records, balancing GC content.

1. Introduction

The healthcare domain in India is still very much dependent on a paperwork-based model. Going to any clinic in Maharashtra would result in encountering piles of printed lab test reports, prescriptions written on paper, and history of patients stored in files that might not be accessed for years to come. Even when the patient record is digitalized, it is stored in PDF format without being compressed or indexed. In the

initial stages of our research, we met with and discussed with two practicing doctors – one practicing Homoeopathy at Mangalmurti Clinic in Pen, and the other practicing Ayurveda at Dr. Mithilesh Patil's Clinic. Both the doctors agreed to share their views on the problem of huge volumes of data stored in an unreadable form.

It goes beyond mere problems of storage. There is long-term value in keeping clinical records, since a patient's haematological parameters recorded several years ago will still have clinical relevance, and Ayurveda's constitution records are designed to be long-term and longitudinal. Without proper archiving, these records lose their value over time. Similarly, clinical diagnostics laboratories contribute to the problem by generating a large amount of structured output that is naturally parameterized but is not compressed in semantic terms and is not archived, trapping structured information within an unstructured format.

Concurrently, another area of research has proved that artificial DNA is capable of storing data at far greater densities than conventional magnetic or solid state memory, up to 215 petabytes per gram and with long-term stability for centuries if stored properly [16-20]. Fundamental work by Church et al. [16], Goldman et al. [17], Organick et al. [18], and Erlich and Zielinski [15] have confirmed that binary information can be effectively and stably encoded within nucleotide sequences. These methods were later improved upon through the development of systems like DNA Fountain [15], DNA-Aeon [6], and spatial encoding methods [5] pushed storage density close to the theoretical maximum of two bits per nucleotide.

In almost all cases, what constrains the existing DNA-based storage pipelines is the fact that they treat their input as an unstructured binary sequence. The data arrives as bytes, is transformed into the four-symbol alphabet {A,T,C,G}, and is subsequently decoded into bytes. There is no consideration for semantics at all; the range limits on which a haemoglobin level test must always lie, and the fixed structure of a complete blood count report regardless of whether thousands of people are screened – the nature of the data itself is ignored.

These considerations are especially significant for clinical records data, as clinical records are far more structured than any other type of data. They contain values limited to numeric ranges, values taken from a finite set of categories, a repeatable template structure for multiple record entries, and relationships among fields that can be predicted using statistics. For example, a patient's haemoglobin levels and their red blood cell count have a correlation with one another. In addition, even if two medical records are from different branches of medicine, they would still have structure to themselves.

This paper discusses the architectural aspects of this problem. The design, encoding approach, and feasibility analysis for Project Helix – an architectural design for the archival of clinical data through synthesis into artificial DNA strands – are outlined. This project consists of modeling of medical records as schema-based representations, semantic normalization with respect to physiological parameters, compression using a layered classical and arithmetic coding approach, and mapping into artificial DNA strands within the constraints of biochemical requirements throughout the encoding process.

The scope of this paper is deliberately limited to the architectural layer of the proposed framework. Additional AI-driven latent data compression and intelligent reconstruction layer will be presented in follow-up papers to ensure clarity and self-containment of our contribution.

The primary contributions of this paper can be summarized as follows:

1. A schema-based model for representing heterogeneous medical records from Allopathic, Ayurvedic, Homoeopathic, and laboratory data sets.
2. A semantic normalization algorithm that utilizes bounded physiological parameter values to reduce entropy prior to statistical compression.

3. An approach to layered classical and arithmetic compression (Run-Length Encoding, Huffman Coding, Arithmetic Coding).
4. A DNA encoding technique with constraint awareness, ensuring GC-content equilibrium and homopolymer length requirements are fulfilled as part of nucleotide mapping, without any additional encoding post-processing.
5. A feasibility study of the developed prototype, involving progress in compression ratio and biological constraint satisfaction by various encoding techniques.

The rest of this article is arranged as follows. Section 2 provides an overview of the evolution of DNA data storage in light of this research. Section 3 analyzes the relevance of the evolution for medical record storage applications. Section 4 outlines the knowledge gap motivating the development of the suggested model. Section 5 highlights the challenges involved in encoding clinical data into DNA. Section 6 provides a literature review on the subject with a comparison of existing solutions. Section 7 introduces the proposed architecture. Section 8 describes the reasoning behind the architecture and its importance. Section 9 considers the issues associated with the implementation and feasibility of the model.

2. Evolution of DNA-Based Data Storage

2.1 Early Feasibility Demonstrations

The potential of DNA as a reliable archival material became evident when Church, Gao, and Kosuri [20] showed that arbitrary digital data can indeed be directly encoded in synthesized oligonucleotides. For the first time, these researchers provided proof that arbitrary content in the size of an entire book can be encoded and successfully decoded from synthetic DNA based on the principle of base pair substitution. Subsequently, Goldman et al. [17] took a step further, proposing an encoding scheme which uses redundancy for robustness against sequencing errors, something previous methods could not address. Finally, Grass et al. [18] focused their efforts on the problem of longevity and proved the long-term stability of DNA encoded into silica microspheres under accelerated aging conditions, corresponding to centuries of natural storage.

Overall, these initial demonstrations served as proof of principle for two distinct, yet related, aspects of DNA storage – the ability to store vast amounts of information per unit of physical volume, as well as longevity exceeding those offered by all other forms of archiving technology. What these studies did not address, however, was the efficiency of binary representation before encoding.

2.2 Density-Oriented Encoding Schemes

The second wave of work aimed at pushing capacity toward its maximum possible value of two bits per nucleotide. The DNA Fountain by Erlich and Zielinski [15] used the principle of fountain codes, which have been widely employed for tolerating packet losses during erasure-tolerant communication over the network, while providing close-to-optimal information capacity per nucleotide, reaching 1.57 bits per nucleotide and possessing high error correction capability, being tolerant to the loss or corruption of a significant portion of sequencing results. Hybrid mapping and repeat accumulate schemes similarly reached almost maximum DNA channel capacity by means of complex channel coding methods with explicit consideration of DNA channel as an erasure channel.

Spatial encodings allowed encoding information using a two-dimensional nucleotide matrix, where the amount of bits per nucleotide approached three by using positionally based encoding rather than sequential. DNA-Aeon [6, 8] utilized arithmetic coding combined with the idea of constraint satisfaction

done within one iteration; this work was quite inspirational as regards this study, because it proved that the problem of data compression does not have to come before enforcing biological constraints sequentially.

2.3Application-Specific Extensions

However, more recent studies have started to optimize DNA storage according to the types of contents to which they apply, as opposed to using content-agnostic encoding approaches for any type of input. For instance, progressive image storage systems, like PIC-DNA [9], used layered and selective decoding for multimedia contents, where low resolution previews can be obtained without decoding the whole sequence — thereby proving that designing encoding techniques based on their specific content could provide significant improvements compared to content-agnostic approaches. Moreover, tokenized approaches for videos have been considered to exploit temporal redundancies before encoding to DNA.

The above progression, from general encoding to content-aware approaches, has motivated the choice of a certain type of content, namely clinical data, which has not yet been considered in previous research even though it might have a greater potential than the other mentioned data types.

3. Relevance to Healthcare Data Archival

3.1Clinical Data as a Distinct Storage Class

In contrast to the general data employed to test the efficiency of any DNA data storage technique, healthcare data exhibit particularities that distinguish them and which can be exploited through specialized encodings. Lab measurements are restricted within certain limits determined by physical constraints. Diagnostic terms are selected from a relatively small pool of standardized vocabulary. Moreover, forms used by clinics exhibit a significant degree of repetitiveness when applied to the patient population. This is an inherent feature of data we observed in our preliminary consultation sessions with the Homoeopathy and Ayurveda doctors (see Section 1). Indeed, hemoglobin level, for instance, rarely deviates from the interval [5,25] g/dL.

3.2Multi-System Documentation Variability

Another issue related to the Indian healthcare scenario is that of various formats of documentation of medical information. In the case of allopathic lab reports, they are largely numerical and highly dependent on parameters for each of the tests like blood counts and lipids. The Ayurvedic consultations involve longitudinal documentation of qualitative descriptions of constitution along with numbers, including body types and history of herbals over an extensive period of time. In the homoeopathic format of medical data recording, individualized symptom-based information, including emotional behavior, are documented, without a field equivalent to these in allopathic records. An archive designed for implementation in a practical clinical environment requires flexibility to cater to such variations in documentation style, which forms the basis of the structure modeling approach mentioned in Section 7.

3.3Long-Term Preservation Requirements

Not like several other data forms where archival is simply an inactive process, the clinical data remains relevant even years down the line—a factor that matches perfectly with the stability and longevity of DNA molecules for hundreds of years [18], yet is hardly achieved by modern encoding techniques that completely overlook the intrinsic structures of the data while compressing it. The information stored in a patient's record today may be accessed and decoded even decades from now, when the actual device

storing this record or its original format does not even exist anymore; it can only be achieved with the help of DNA and proper encoding technique.

4. Research Gaps

The gaps in research identified via synthesizing Sections 2 and 3 with respect to our findings from the clinical consultations can be summarized as follows:

- No semantic structuring step before compression. In current DNA storage systems, health records are sent through the compressing function as unstructured strings of binary digits, ignoring the fact that laboratory values, diagnostic codes, and treatments all follow bounded and predictable distributions.
- No modeling based on domain knowledge. Current generic encoding systems are unable to handle the differences in format between Allopathic, Ayurvedic, Homoeopathic, and diagnostic laboratory medical records discovered by us.
- Optimization performed without consideration for information loss. While encoding techniques like DNA Fountain and spatial encoding as well as repeat-accumulate codes [11, 12, 15] focus on optimizing the number of required nucleotides, entropy reduction comes separately.
- Biological limitations treated as post-compression concerns. Biological constraints such as GC-content adjustment and homopolymer avoidance appear as post-compressing steps in the vast majority of existing pipelines.
- No single architectural approach to healthcare archiving. To our knowledge, no previous work exists that considers schema-based modeling, normalization of the domain knowledge, and encoding with biological limitations into one pipeline.

All these gaps can only lead to one conclusion: great strides have been made within the field towards optimization in the process of encoding the data into the DNA, but what needs to be done to the data prior to any encoding has not yet been explored. This becomes the primary rationale behind the design presented in Section 7

5. Challenges in Structured Clinical Data Encoding

The choice of designing the encoding pipeline based on structured clinical data rather than raw generic binary data is bound to lead to a number of issues which will form the basis of some of the design decisions made for the system as discussed in Section 7.

Incongruent documentation schema. Records in allopathic medicine, Ayurvedic medicine, Homoeopathic medicine, and laboratory reports each have their own internal structures, necessitating that the modelling phase should be robust enough to handle all the above without reducing the differences between the specialties through normalization.

Loss-less compression required to preserve reversibility. As the clinical data cannot have any uncontrolled loss of information, all phases of compression have to be reversible, hence restricting the coding schemes used to only deterministic and not approximative or lossy.

Compression vs. biological synthesizability. Over-compression ignoring the GC-content and homopolymer constraints can generate highly compressed binary strings which will generate unstable/unsynthesizable DNA strings, hence joint instead of sequential optimization of the two aspects.

Drifting of schema over time even among records from the same specialty. There is always variation in the schema employed by doctors/lab software even when considering records from the same medical specialty.

Demonstrating biological feasibility without real synthesis. Since this is done using just software simulation, the validation had to rely on constraints checking (GC content, maximum homopolymer size) rather than actual synthesis, which is considered one limitation of this evaluation and is further addressed in Section 9.

6. Related Works

Table 1 summarizes the most directly relevant prior systems, the techniques they employ, and the specific gap each leaves with respect to structured, healthcare-oriented archival.

Table 1. Comparative summary of related DNA storage approaches.

Work	Technique / Focus	Gap Relative to This Work
Taluja & Gupta	Survey of binary-to-DNA mapping strategies	General description only; no semantic compression or implementation framework
Arunachalam & Arunachalam	Index-based RLE + Huffman for genomic sequences	Built for biologic sequences, not artificial encoding of structured documents
Chen et al. (Repeat-Accumulate)	Hybrid mapping achieving ~1.98 bits/nt	Density-based; no semantic or domain-aware pre-processing
DNA Fountain [15]	Fountain coding for dropout tolerance	Domain-agnostic; takes input as generic binary string
Spatial Encoding [5]	3 bits/base via spatial mapping, fully reversible	No domain-specific structure; emphasis only on efficient mapping
DNA-Aeon [6, 8]	Arithmetic coding with inline constraint handling	Generic binary coding; not any application of structured domain
PIC-DNA [9]	Layered progressive compression for images	Specifically for images; cannot be applied to structured data
This work (Project Helix)	Schema modelling + semantic normalization + inline constraint-aware encoding	Targeted at structured clinical records of multiple systems

There are two recurring trends within the literature surveyed above. First, compression and DNA-constraint satisfaction are considered sequential and separate processes, with a compressor compressing the binary stream and then the result being validated for DNA feasibility, as opposed to being optimized together. Second, there is no mechanism to account for the semantic structure of clinical documents, as opposed to the structure of images, even in content-aware schemes like PIC-DNA. The architecture presented in Section 7 aims directly at remedying both these shortcomings.

It is also relevant to point out the adjacent area of homomorphic encryption [14], which is orthogonal to the current discussion and does not intersect with any of the prior literature surveyed above. Although this field of study is not the main focus of this paper, it may be seen as a natural follow-up direction from the proposed architecture, as discussed in Section 10 below.

7. Proposed Architecture

7.1 Architecture Overview

The proposed system model consists of the following five sequential operations performed on structured clinical data: structured data modelling, semantic normalization, classical multi-layer compression, DNA encoding based on constraints, and intelligent retrieval. Entropy reduction or biological feasibility enforcement does not exclude reversibility in any of these steps. The entire proposed system is shown in

7.2 Structured Clinical Data Modelling

Unlike the conventional approach that compresses PDF files as is, our methodology involves extraction of clinically significant content from such PDF documents, namely, numerical parameters, categorizations, and treatment data. The extracted content is then converted into a standardized representation according to a predefined schema, while discarding any non-essential formattings such as fonts, layout design, and repetition of headers. Due to the fact that medical documentation of allopathy, ayurveda, homoeopathy, and laboratories have their own standards of writing down clinical data, we have created an adaptable schema where only specialization-related parameters are included without transforming all inputs into a universal numeric schema. All three types of fields, which include numerical parameters, categorizations, and ordinal data, are separated in order to be normalized individually.

7.3 Semantic Normalization

After structuring, the process carries out domain-specific normalization, which minimizes the amount of entropy prior to compression based on statistics. In the normalization phase, there are four tasks accomplished: range-based encoding of physiologic data, based on the fact that physiologic data like levels of haemoglobin or platelets occur in ranges, making it possible to encode them in a way that requires less number of bits compared to numeric coding; elimination of unnecessary units of measure that are inherent in the schema; symbol table-based encoding of the categorical variables that use domain vocabularies, which assigns short codes to frequently recurring diagnostic descriptions; and schema-based structural normalization to ensure consistent format regardless of specialties.

7.4 Multi-Layer Classical Compression

The normalized information is processed in two stages of compression. The first stage utilizes Run-Length Encoding as a means to take advantage of repetitive schema design, especially useful considering the presence of those structures that appear repeatedly in the dataset across patients within a clinic; then follows Huffman encoding, wherein shorter codes are assigned to the symbols based on their probabilities, using the clinical record training dataset; all these operations are deterministic and can be reversed precisely. The next stage utilizes arithmetic coding technique, which compresses the entire stream of normalized information in the form of a probability range as opposed to symbol-by-symbol processing, and achieves higher compression efficiency especially if there are interdependencies in adjacent schema elements, as was discussed in Section 1.

7.5 Constraint-Aware DNA Encoding

The compressed binary stream is mapped to a DNA sequence consisting of the four nucleotides: adenine (A), thymine (T), cytosine (C), and guanine (G), where the constraints are applied at runtime as the process of mapping takes place instead of being a corrective step taken after the mapping. The encoding process ensures that the following constraints be met: GC content should be between 40% and 60%, as sequences with an out-of-bounds GC percentage result in low synthesis efficiency and poor sequencing accuracy; no homopolymers longer than two consecutive nucleotides; and deterministic one-to-one mapping, which assures that the original binary string can be recovered without using any auxiliary look-up table.

7.6 Retrieval Pathway

During the retrieval process, the encoded sequence of DNA becomes decoded to go through the exact inverse operations to obtain a structured, normalized representation of patient data. This process, for the purpose of the architecture defined in this paper, is deterministic in nature; the learning and intelligence-based reconstruction capability of the framework, however, is defined separately in Section 1 and involves the use of a model to reconstruct structured fields from a latent form.

8. Architectural Significance and Design Rationale

8.1 Why Semantic Modelling Precedes Compression

Putting semantic normalization and structured data modeling in front of statistical compression, instead of just compressing raw file content, is the critical architectural decision here. This makes entropy reduction possible at the level of information, using known limits on physiology and repetitive schema, prior to any generic compression algorithm being used, as opposed to trying to reduce entropy through statistics alone on an unstructured byte stream. The impact of this sequence, as presented in Section 9, demonstrates that there is indeed a quantifiable benefit from semantic normalization itself prior to any classical encoding procedure..

8.2 Why Constraints Are Enforced Inline

Incorporation of these two constraints into the nucleotide encoder itself, as opposed to employing the corrective passes in a separate step after the data encoding, eliminates the necessity for any form of redundancy padding that would come as an overhead due to such corrections. It has been motivated by the joint source-constraint coding approach used in DNA-Aeon [6, 8] but adapted specifically to the properties of the preprocessed stream as opposed to arbitrary binary input. The fact that the input stream to this module has already undergone entropy reduction makes for a smaller search space within the constraint-checking algorithm compared to a generic compressor's output.

8.3 Why the Schema Layer Is Specialty-Adaptable

However, the rationale behind keeping the specialization-dependent properties in the schema instead of normalizing all input to a single, numeric, format comes directly out of the clinical discussions presented in Section 1. A design meant for practical application in Allopathic, Ayurvedic, and Homoeopathic medicine needs to take account of the different structures of these kinds of medical records instead of discarding any data that does not match a common input format. And indeed, it is this very requirement of being specialized for the healthcare industry that makes it impossible to just apply any generic DNA compression system straight away; not because it is unable to compress efficiently, but because it is unaware of the domain's specifics.

8.4 Extensibility Toward Intelligent Reconstruction

Given the clean architecture where the compression stage (Section 7.4) is separate from the encoding stage of biology (Section 7.5), the system is architected in such a way as to ensure that the classic compression stage may be enhanced by learning the latent space without affecting the constraint-aware encoder process. It is because of this separation that the AI-driven reconstruction extension proposed in Section 1 may leverage this architecture without requiring a rewrite; the constraint-aware encoder process (Section 7.5) would simply treat a latent space compressed binary stream the same way it processes an arithmetic coded one.

9. Implementation Challenges and Feasibility Results

The process of implementing and validating the prototype revealed several practical challenges that merit consideration in addition to the feasibility findings achieved using our implemented pipeline.

9.1 Practical Implementation Challenges

Input schema flexibility for multiple specialties. A single, general-purpose schema that is able to describe Allopathic lab values, Ayurvedic constitutions, and Homoeopathic cases was designed and refined using the sample medical records gathered through collaborations with Mangalmurti Clinic and Dr. Mithilesh Patil's clinic.

Independence of reversible encoding stages. Since lossless compression cannot introduce data corruption, each encoding step had to be confirmed to produce a reversible output relative to the structure of the input record, independently tested before integrating it into the complete pipeline.

Trade-off between correction and density. Using a max-run-length of two, as opposed to the more common run-length of three in other existing schemes, decreased the number of correction stages, but with a slight penalty in terms of compression efficiency.

Small dataset size. While some preliminary feasibility findings are provided here, they are based on a relatively small dataset of structured clinical records, which were obtained as part of our collaboration with hospitals, further testing with larger data will be covered in Section 10.

No DNA Synthesis in vitro. Any biological plausibility claims in this article have been made based on the computationally verified constraint satisfaction, i.e., the GC-content range and homopolymer runs, but not on any confirmation from the synthesis of real oligonucleotides.

On semantic normalization alone, the compression rate already improved by 1.33x even before any general-purpose compressor is used, confirming the architectural idea in Section 8.1 that exploiting structure in advance of compression yields independent and additive improvements over classical statistical coding. The constraint-aware encoding algorithm outperformed DNA Fountain in terms of GC-content variability, with the former exhibiting standard deviation of 3.1% versus 4.2% for the latter, and limiting the length of homopolymer sequences to two, falling below three which is considered a high risk of synthesis errors in the literature [6]. Each step of the entire compression procedure, including decompression, produced an identical representation of the structure of every test record without exception, with no bit-level difference between input and output found in any test case.

This outcome, despite being produced using the prototype, validates the key architectural assumption of this paper: that domain-aware compression, being orthogonal to classical statistical compression, can be used to complement rather than duplicate the effect; moreover, that biological constraints can be satisfied without losing compression effectiveness when constraints are satisfied inline instead of post-compression.

10. Conclusion and Future Work

This paper detailed the design and constraint-sensitive encoding strategy for Project Helix, an artificial DNA archival technique specially designed for structured, heterogeneous clinical records. This work bridges the gap left by DNA storage research literature to date by ensuring that semantically aware organization and domain-sensitive semantics precede classical and arithmetic compression techniques and ensuring biological feasibility constraints through the use of inline GC-content and homopolymer constraints during nucleotide mapping, as opposed to a decoupled process of compression followed by correction.

Our implemented prototype obtained a 2.53× compression ratio when compared to structured binary data utilizing only classical and arithmetic compression methods, with the DNA sequences created having met the biological constraints – 48.1% GC-content and two homopolymers max. – surpassing binary mapping and achieving comparable stability to that of DNA Fountain.

The contributions made in this paper have been consciously restricted to the architectural and constraint-based encoding levels of the above-mentioned framework. The classical compression techniques, as outlined in Section 7.4, are inherently designed, as elaborated in Section 8.4, to be extendible to an additional level of latent AI-based compression without changes in the architecture. Together with the intelligent reconstruction of data, homomorphic encryption for privacy-preserving archival (Section 6), and large-scale evaluation using real hospital data sets, the above-mentioned components lay the groundwork for future research along those lines.

However, as far as the main contribution of this work is concerned, our stance can be summed up as follows: Instead of considering how better to pack binary data into DNA, this work explores the issue of what should be done with healthcare data before encoding it into DNA. As such, we believe that the approach suggested above represents a fruitful direction in the development of a truly useful DNA archival system for preserving long-term healthcare data.

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Disclosure of Interests.

The authors have no competing interests to declare that are relevant to the content of this article.

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